A circular loop of radius 20 cm and resistance 2 Ω is placed in a time varying magnetic field $\vec{B} = (2t^2 + 2t + 3)$ T. At t = 0, for the plane of the loop being perpendicular to the magnetic field and, the induced current in the loop at $t = 3$ s is $\frac{\alpha}{50}$ A. The value of α is _____.

(Take $\pi = 22/7$)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69112141
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112141
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 691121501 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

What volume of hydrogen gas at STP would be liberated by action of 50 mL of H_2SO_4 of 50% purity (density = 1.3 g mL^{-1}) on 20 g of zinc ?

Given : Molar mass of H, O, S, Zn are 1, 16, 32, 65 g mol^{-1} respectively.

Options :

6911211701. 5.824 L

6911211702. 7.428 L

6911211703. 6.892 L

6911211704. 8.375 L

Question Number : 52 Question Id : 691121502 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following statement(s) is/are **true** ?

A. If two orbitals have the same value of $(n+l)$, the orbital with lower value of n will have lower energy.

B. Energies of the orbitals in the same subshell increase with increase in atomic number.

C. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

D. Among 5f, 6s, 4d, 5p and 5d orbitals, none of the orbitals have 2 radial nodes.

Choose the **correct** answer from the options given below :

Options :

6911211705. A, B and C only

6911211706. A and C only

6911211707. C and D only

6911211708. A only

Question Number : 53 Question Id : 691121503 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The covalent radii of atoms A and B are r_A and r_B, respectively. The covalent bond length and total length of AB molecule are respectively

Options :

6911211709. $(r_A + r_B),\ 2(r_A + r_B)$

6911211710. $\frac{1}{2}(r_A + r_B),\ (r_A + r_B)$

6911211711. $(r_A + r_B),\ (r_A + r_B)$

6911211712. $2(r_A + r_B),\ \frac{1}{2}(r_A + r_B)$

Question Number : 54 Question Id : 691121504 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following data for the reaction

$X_2(g) + Y_2(g) \rightleftharpoons 2XY(g)$

at 600 K. The $\Delta_r G^{\ominus}$ (in kJ mol^{-1}) for the reaction is :

Compound	$\Delta_f H^{\ominus}_{600K}$ (kJ mol^{-1})	$S^{\ominus}_{600K}$ (J mol^{-1} K^{-1})
XY(g)	42	200
$X_2(g)$	8	140
$Y_2(g)$	80	250

Options :

6911211713. -21000

6911211714. -10

6911211715. -1000

6911211716. -9.012

Question Number : 55 Question Id : 691121505 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The **correct** order of molar heat capacities measured at 298 K and 1 bar is :

Options :

6911211717. Copper(s) > Bromine(l) > Helium(g)

6911211718. Bromine(l) > Copper(s) > Helium(g)

6911211719. Helium(g) > Bromine(l) > Copper(s)

6911211720. Helium(g) > Bromine(l) = Copper(s)

Question Number : 56 Question Id : 691121506 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The reaction $A(g) \rightleftharpoons B(g) + C(g)$ was initiated with the amount 'a' of A(g). At equilibrium it is found that the amount of A(g) remaining is $(a - x)$ at a total pressure of p.

The equilibrium constant Kp of the reaction can be calculated from the expression :

Options :

6911211721. $\frac{x^2}{a^2 + x^2} \times p$

6911211722. $\frac{x^2}{a^2-x^2}\times p$

6911211723. $\frac{a+x^2}{x^2}\times p$

6911211724. $\frac{a^2-x^2}{x^2}\times p$

Question Number : 57 Question Id : 691121507 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

One half cell in a voltaic cell is constructed by dipping silver rod in $AgNO_3$ solution of unknown concentration, other half cell is Zn rod dipped in 1 molar solution of $ZnSO_4$.

A voltage of 1.60 V is measured at 298 K for this cell. What is the concentration of Ag^+ ions used in terms of log x ($x = [Ag^+]$) ?

$$E^{\ominus}_{Zn^{2+}/Zn} = -0.76V, \quad E^{\ominus}_{Ag^+/Ag} = +0.80V, \frac{2.303RT}{F} = 0.059\,V$$

Options :

6911211725. $\frac{2}{3.9}$

6911211726. $\frac{4}{5.9}$

6911211727. $\frac{2.9}{2}$

6911211728. $\frac{5.9}{4}$

Question Number : 58 Question Id : 691121508 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : The number of pairs among [Al_2O_3, Cr_2O_3], [Cl_2O_7, Mn_2O_7], [Na_2O, V_2O_3] and [CO, N_2O] that contain oxides of same nature (acidic, basic, neutral or amphoteric) is 4.

Statement II : Among Na_2O, Al_2O_3, CO and Cl_2O_7, the most basic and acidic oxides are Na_2O and Cl_2O_7, respectively.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211729. Both **Statement I** and **Statement II** are true

6911211730. Both **Statement I** and **Statement II** are false

6911211731. **Statement I** is true but **Statement II** is false

6911211732. **Statement I** is false but **Statement II** is true

Question Number : 59 Question Id : 691121509 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Aluminium upon reaction with NaOH forms $[Al(OH)_6]^{3-}$ ion.

Statement II : The geometry of ICl_4^-, ClO_3^- and IBr_2^- is square planar, pyramidal and linear respectively.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211733. Both **Statement I** and **Statement II** are true

6911211734. Both **Statement I** and **Statement II** are false

6911211735. **Statement I** is true but **Statement II** is false

6911211736. **Statement I** is false but **Statement II** is true

Question Number : 60 Question Id : 691121510 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Presence of large number of unpaired electrons in transition metal atoms results in higher enthalpies of their atomisation.

Statement II : $d_{xy} = d_{xz} = d_{yz} < d_{x^2-y^2} = d_{z^2}$ and $d_{x^2-y^2} = d_{z^2} < d_{xy} = d_{xz} = d_{yz}$ are the d-orbital splittings in $[Fe(H_2O)_6]^{3+}$ and $[Ni(Cl)_4]^{2-}$ complex ions respectively.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211737. Both **Statement I** and **Statement II** are correct

6911211738. Both **Statement I** and **Statement II** are incorrect

6911211739. **Statement I** is correct but **Statement II** is incorrect

6911211740. **Statement I** is incorrect but **Statement II** is correct

Question Number : 61 Question Id : 691121511 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the **correct** statements from the following

A. $[Fe(C_2O_4)_3]^{3-}$ is the most stable complex among $[Fe(OH)_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(SCN)_6]^{3-}$

B. The stability of $[Cu(NH_3)_4]^{2+}$ is greater than that of $[Cu(en)_2]^{2+}$

C. The hybridization of Fe in $K_4[Fe(CN)_6]$ is d^2sp^3

D. $[Fe(NO_2)_3Cl_3]^{3-}$ exhibits linkage isomerism

E. NO_2^- and SCN^- ligands are NOT ambidentate ligands

Choose the **correct** answer from the options given below :

Options :

6911211741. A, B, C, D and E

6911211742. B, C and D only

6911211743. A, C and D only

6911211744. A, C and E only

Question Number : 62 Question Id : 691121512 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I **Purification technique**		**List - II** **Used to separate**
A.	Simple distillation	I.	Steam volatile compound
B.	Fractional distillation	II.	Two liquids with large difference in boiling points
C.	Steam distillation	III.	Liquid decomposing at its boiling point
D.	Distillation under reduced pressure	IV.	Two liquids with close boiling points

Choose the **correct** answer from the options given below :

Options :

6911211745. A-II, B-III, C-I, D-IV

6911211746. A-II, B-IV, C-I, D-III

6911211747. A-II, B-IV, C-III, D-I

6911211748. A-IV, B-III, C-II, D-I

Question Number : 63 Question Id : 691121513 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

IUPAC name of the some alkenes are given below.

Find out the correct stability order.

A. 2-Methylbut-2-ene

B. *cis*-But-2-ene

C. 2, 3-Dimethylbut-2-ene

D. Prop-1-ene

Choose the **correct** answer from the options given below :

Options :

6911211749. C > A > B > D

6911211750. C > A > D > B

6911211751. B > D > A > C

6911211752. A > B > C > D

Question Number : 64 Question Id : 691121514 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass 72 g mol^{-1}.

Options :

6911211753. 1, 1 - Dimethylcyclopropane

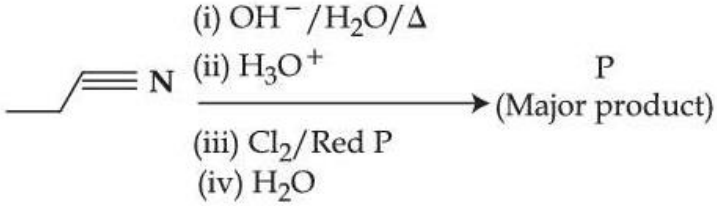
6911211754. 2, 2 - Dimethylpropane

6911211755. 2 - Methylbutane

6911211756. n-pentane

Question Number : 65 Question Id : 691121515 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Complete the following reaction sequence and give the name of major product 'P'.



Options :

6911211757. 2-Chloropropanoic acid

6911211758. 3-Chloropropanoic acid

6911211759. 1-Chloropropane

6911211760. 2-Chloropropane

Question Number : 66 Question Id : 691121516 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : The condensation reaction between $CH_3-CH=O$ and $H_2N-\underset{\underset{H}{|}}{N}-\overset{\overset{O}{||}}{C}-NH_2$ under optimum pH will produce $CH_3-CH=N-\overset{\overset{O}{||}}{C}-\overset{\overset{H}{|}}{N}-NH_2$

Statement II : The molecule, $Ph-CH\begin{matrix}\diagup O-H\\ \diagdown O-CH_3\end{matrix}$ will generate $Ph-CH=O$ in the presence of dilute acid.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211761. Both **Statement I** and **Statement II** are true

6911211762. Both **Statement I** and **Statement II** are false

6911211763. **Statement I** is true but **Statement II** is false

6911211764. **Statement I** is false but **Statement II** is true

Question Number : 67 Question Id : 691121517 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Heating benzamide with bromine in an ethanolic solution of sodium hydroxide will give benzylamine.

Statement II : Nitration of aniline with HNO_3/H_2SO_4 at 288 K produces *m*-nitroaniline in higher amount than *o*-nitroaniline (pH adjusted).

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211765. Both **Statement I** and **Statement II** are true

6911211766. Both **Statement I** and **Statement II** are false

6911211767. **Statement I** is true but **Statement II** is false

6911211768. **Statement I** is false but **Statement II** is true

Question Number : 68 Question Id : 691121518 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the **incorrect** statement about tertiary structure of proteins.

Options :

6911211769. They can be fibrous or globular in structure.

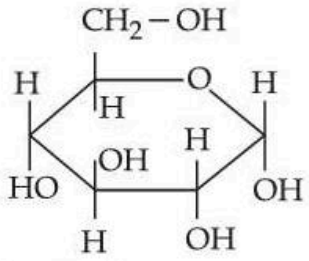
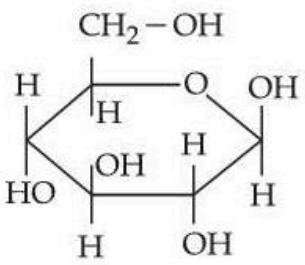
6911211770.
The main forces that stabilize the structure are hydrogen bonding, disulphide links, van der Waals and electrostatic forces of attraction.

6911211771. The structure remains intact when exposed to pH changes.

6911211772.
A linear polypeptide chain will convert to a secondary structure and then further folding of the secondary structure will convert to tertiary structure.

Question Number : 69 Question Id : 691121519 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I :  and  are two anomers of D-(+)-glucose.

Statement II : The open chain forms of D-glucose and D- fructose contain three similar chiral carbons at C_3, C_4 and C_5.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211773. Both **Statement I** and **Statement II** are true

6911211774. Both **Statement I** and **Statement II** are false

6911211775. **Statement I** is true but **Statement II** is false

6911211776. **Statement I** is false but **Statement II** is true

Question Number : 70 Question Id : 691121520 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A paper dipped in a dil. H_2SO_4 solution of 'X' upon treatment with SO_2 gas turns into green. The compound 'X' is :

Options :

6911211777. KI-starch

6911211778. $KMnO_4$

6911211779. $Pb(CH_3COO)_2$

6911211780. $K_2Cr_2O_7$

Chemistry Section B

Section Id :	69112142
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112142
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 691121521 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The total number of unpaired electrons present in the d^3, d^4 (low spin) d^5 (high spin), d^6 (high spin) and d^7 (low spin) octahedral complex systems is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 691121522 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

RMgI when treated with ice cold water liberated a gas which occupied 1.4 dm^3/g at STP. The gas produced is further reacted with iodine in presence of HIO_3 to give compound (X). Compound (X) in presence of Na and dry ether produced compound (Y). Molar mass of compound (Y) is _____ $g\ mol^{-1}$. (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 73 Question Id : 691121523 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

20 g hemoglobin in a 1 L aqueous solution (A) at 300 K is separated from pure water by semi permeable membrane. At equilibrium the height of solution in a tube dipped in a solution (A) is found to be 80.0 mm higher than the tube dipped in water.

The molar mass of hemoglobin is _____ kg mol^{-1}. (Nearest integer)

(Given : g = 10 m s^{-2}, R = 8.3 kPa dm^3 $K^{-1}mol^{-1}$, density of solution = 1000 kg m^{-3})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 691121524 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

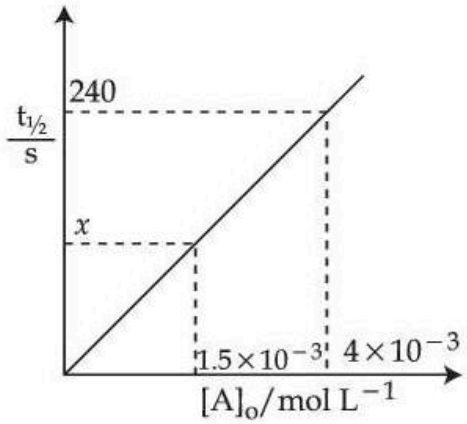
At 298 K, the molar conductivity of x% (w/w) MX solution (aqueous) is 123.5 S cm^2 mol^{-1}. The conductance of same solution is 1.9×10^{-3} S. The value of x is _____ $\times 10^{-2}$.

(Given : cell constant = 1.3 cm^{-1}; molar mass of MX is 75 g mol^{-1}, density of aqueous solution of MX at 298 K is 1.0 g mL^{-1})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 691121525 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For a reaction A → P at T K, the half life ($t_{1/2}$) is plotted as a function of initial concentration $[A]_o$ of A as given below.



The value of x in the given figure is _____s (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1